

Active Recall Self-Quiz Bank

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The Science of Learning SQL — Chapter-by-Chapter Question Bank

How to use this bank: Close your notes. Write or type your answer from memory before checking. Passive re-reading feels productive but builds no durable skill — the forgetting curve will erase it within days. Reconstruction is the mechanism that rewires technical memory.

Spaced Repetition Schedule

Session	Chapters to Quiz	Days Since Last Review
Day 1	Ch. 3	— (first exposure)
Day 2	Ch. 3 + Ch. 4	1
Day 4	Ch. 3, 4 + Ch. 5	2
Day 7	Ch. 3–5 + Ch. 6	3
Day 12	Ch. 3–6 + Ch. 7	5
Day 20	Full bank: Ch. 3–7	8

Return to questions that stumped you at each session. Struggle during retrieval is the signal that durable learning is occurring.

Chapter 1 — The Cognitive Science Foundation

1. Name the two learning mechanisms that Chapter 1 identifies as building lasting SQL fluency, and explain in one sentence why each works.
 2. What does the forgetting curve predict will happen to SQL syntax you read once but never actively retrieve?
 3. Describe the "pyramid" metaphor for progressive skill layering. Why does skipping foundational concepts cause cognitive overload later?
 4. What is the difference between recognising a SQL keyword and being able to reconstruct it? Which one do professional analysts actually need?
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Chapter 3 — Your First SELECT Statement

5. Write from memory the minimal SQL statement that retrieves every column from a table called customers.
 6. Rewrite that query so it retrieves only customer_id and email.
 7. A colleague's result set contains 47 rows but several city names repeat. Write the query that returns only unique city values from a customers table.
 8. Your output column is labelled annual_revenue. Write the clause that renames it to Annual Revenue in the results without altering the table.
 9. You run a query and receive an error. According to Chapter 3, what should your first instinct be — and why are error messages described as "diagnostic feedback" rather than failures?
 10. What is the difference between SELECT and listing specific columns? Name one professional reason to avoid SELECT in production queries.
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Chapter 4 — Filtering Rows With WHERE

11. Write a query that retrieves all rows from orders where status equals 'shipped'.
 12. Extend that query to also require that order_total is greater than 500.
 13. A column called delivery_date contains some empty values. Write the condition that returns only rows where delivery_date has no value recorded.
 14. Why can't you write WHERE delivery_date = NULL? What happens when you try?
 15. Write a WHERE clause that returns customers whose last name starts with the letter M. Which operator and wildcard character do you need?
 16. You need orders placed in 2023 only. Write the condition using >= and < on a order_date column.
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Chapter 5 — Sorting and Limiting Results

17. Write a query that returns the 10 most recent rows from an events table sorted by event_date.
 18. Sort products by category ascending, then by price descending within each category. Write the full ORDER BY clause.
 19. List the six standard SQL clauses (SELECT, FROM, WHERE, GROUP BY, HAVING, ORDER BY) in the order you write them, then in the order the database executes them.
 20. What does OFFSET 20 accomplish when combined with LIMIT 10? Describe the practical use case.
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Chapter 6 — Aggregate Functions and GROUP BY

21. Write a query that counts the total number of rows in an orders table.
22. Write a query that returns the average sale_price grouped by product_category.

23. You want only categories where the average sale price exceeds 100. Write the complete query including the correct filter clause.
24. Why does HAVING exist as a separate clause from WHERE? What breaks if you try to filter an aggregate with WHERE?
25. A reviews table has a rating column with some NULL values. What does COUNT(rating) return versus COUNT(*)?
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Chapter 7 — Joins

26. Write an INNER JOIN between orders and customers on customer_id. What rows are excluded from the result?
27. Rewrite that join as a LEFT JOIN. Which table's unmatched rows are preserved?
28. Define a Cartesian product. What mistake in your SQL causes one?
29. A query joins orders, customers, and products. Sketch the two JOIN clauses you need (table names and key columns are your choice — write valid syntax).
30. Why do relational databases split data across multiple tables rather than storing everything in one? Name one benefit.
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Self-Scoring Guide

Score	Interpretation	Action
Wrote it perfectly from memory	Consolidated — schedule next review in 5–8 days	Move forward
Mostly correct, minor syntax gap	Nearly consolidated — review that clause once, re-test tomorrow	Targeted review
Recalled the concept but not syntax	Familiarity without fluency — write the query 3× by hand	Rewrite drill
Could not recall	Not yet encoded — re-read the section, quiz again in 24 hours	Re-exposure